

Yingrui Hou

Machine Learning Data Scientist | Longitudinal Modelling, Validation & Monitoring

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PROFESSIONAL SUMMARY

PhD-trained applied data scientist with several years of experience developing Python/C++ workflows for statistical modelling, machine-learning validation, time-dependent inference, and monitoring on large, noisy real-world measurement datasets.

Experienced in extracting reliable information when ground truth is incomplete, defining robustness checks under dataset mismatch, and translating complex analytical results into documented decisions. Motivated to bring this methodology to longitudinal patient data, real-world evidence, and responsible healthcare machine learning.

CORE SKILLS

- Data science and ML: Python, NumPy, pandas, scikit-learn, statsmodels, CatBoost / Gradient Boosting
- Longitudinal and statistical modelling: time-dependent inference, repeated observations, joint fitting, probabilistic models, maximum-likelihood estimation
- Validation: validation strategy, train/test diagnostics, calibration, bias correction, uncertainty quantification, robustness studies
- Monitoring and data quality: anomaly detection, expected-versus-observed monitoring, dataset-shift diagnostics, root-cause analysis
- Reproducible delivery: Linux, Git, CI/CD practices, modular pipelines, workflow automation, technical documentation
- Technical communication: assumptions and limitations, interpretable results, collaborative review, multidisciplinary teamwork

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Developed reusable Python/C++ analytical pipelines for large, noisy datasets with time-dependent observations, incomplete event-level ground truth, and changing operating conditions.
- Implemented probabilistic and maximum-likelihood models, including joint fits, to estimate latent parameters under background contamination, finite resolution, and sample mismatch.
- Designed validation strategies using control samples, train/test diagnostics, pseudoexperiments, uncertainty stress tests, and stability checks across datasets and selections.
- Built CatBoost-based predictive workflows with feature studies, threshold scans, reproducible scoring, and gradient-boosting reweighting to improve reliability under distribution shift.
- Developed automated monitoring and quality-control workflows that detected anomalies and traced discrepancies to calibration, detector, or software causes.
- Produced documented statistical summaries, diagnostic visualisations, and clear explanations of model assumptions, limitations, and results for multidisciplinary collaborators.
- Worked autonomously across parallel analysis topics in Linux-based shared codebases using Git, modular programming, collaborative review, and reproducible configurations.

Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Developed simulation-backed statistical analyses combining time-resolved signal reconstruction, calibration, probabilistic modelling, and validation against experimental data.

SELECTED OUTCOMES

- Improved reconstructed-energy resolution by about 2x through time-resolved component modelling, statistical fitting, and calibration on realistic measurement data.
- Identified a reconstruction-software issue through structured validation and monitoring; the issue was subsequently corrected downstream.

SELECTED PROJECTS

Time-Series Inference

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

Methods: acceptance modelling, resolution calibration, custom fit components, joint fitting

Domain Adaptation for ML

Built an ML validation and sample-alignment workflow in a shared analysis environment for noisy selection problems under dataset shift.

Methods: CatBoost training, train-test diagnostics, gradient-boosting reweighting

Test-Data Validation & Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics

Advanced Inference Workflow Validation

Contributed the proof-of-principle validation needed to move a new basis-expansion inference idea from theory toward a usable workflow under realistic measurement conditions.

Methods: basis expansion, covariance-aware joint fitting, pseudoexperiment validation, statistical background subtraction

Weak-Signal Inference

Built a reusable C++/Python/ROOT workflow for weak-signal extraction in data where background dominates and event-level truth is unavailable.

Methods: likelihood fitting, control-sample validation, production yield workflows

RESEARCH & KNOWLEDGE SHARING

- Contributed to peer-reviewed publications and internal technical notes covering machine-learning validation, software monitoring, simulation-to-data studies, and statistical inference.
- Turned methodological ideas into reusable workflows with documented assumptions, limitations, validation evidence, and review-ready analytical outputs.
- Evaluated emerging inference methods under realistic measurement effects before their use in collaborative analyses.

EDUCATION & LANGUAGES

- Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021
- BSc in Applied Physics - China University of Mining and Technology, 2010-2014
- Languages: Chinese: Native speaker, English: C1, French: B1 to B2